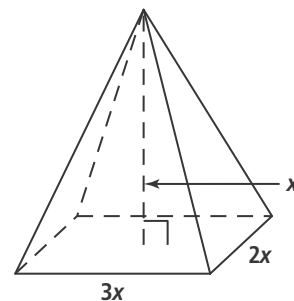


5 Performance Assessment Form B

According to a popular, but untrue, urban legend, oceanographers have discovered images of giant pyramids made of crystal-like substances deep in the ocean. Suppose one of these imaginary pyramids has the dimensions shown at the right.



1. The volume V of a pyramid is $\frac{1}{3}Bh$ where B is the area of the base and h is the height of the pyramid.

Part A

Find an equation that represents V in terms of x . Then find x in terms of V . Write your equation for x in rational exponent form, and rationalize denominators where necessary. Explain.

$$V = 2x^3; B = 3x \cdot 2x = 6x^2 \text{ and } h = x,$$

$$\text{and } V = \frac{1}{3}Bh = \frac{1}{3}(6x^2)(x) = 2x^3.$$

$$x = \frac{(4V)^{\frac{1}{3}}}{2}; V = 2x^3, \text{ so } x^3 = \frac{V}{2}.$$

$$\text{Then } x = \sqrt[3]{\frac{V}{2}} = \frac{\sqrt[3]{V}}{\sqrt[3]{2}} = \frac{\sqrt[3]{V}}{\sqrt[3]{2}} \cdot \frac{\sqrt[3]{2^2}}{\sqrt[3]{2^2}} = \frac{\sqrt[3]{4V}}{2} = \frac{(4V)^{\frac{1}{3}}}{2}.$$

Part B

Suppose the urban legend were true, and a contractor could be hired to raise one of these pyramids and bring it to the surface of the ocean. The estimated cost of the operation $c(x)$ is \$200,000 per meter of pyramid height, with an additional cost $a(V)$ of \$300 per cubic meter of pyramid volume. Suppose a pyramid has a volume of $2,000,000 \text{ m}^3$. What is the height x of the pyramid? What is an equation that represents the total retrieval cost t for a pyramid of any volume with these proportions in terms of x ? How much would it cost to excavate this pyramid? Explain.

$$x = 100 \text{ m}; t(x) = 200,000x + 600x^3; \$620,000,000$$

$$V = 2x^3, \text{ so } 2,000,000 = 2x^3 \text{ and } x = \sqrt[3]{1,000,000} = 100 \text{ m.}$$

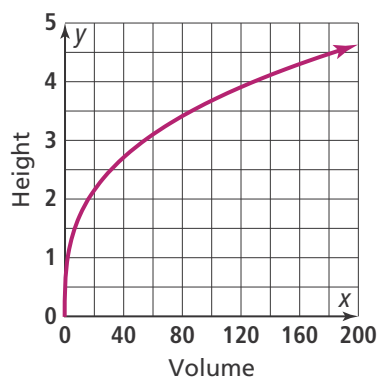
$$a(V) = 300 \cdot V = 300(2x^3) = 600x^3; c(x) = 200,000x$$

$$t(x) = c + a = 200,000x + 600x^3$$

$$\text{For } x = 100 \text{ m, } t(x) = 200,000(100) + 600(100)^3, \\ \text{or } \$620,000,000.$$

2. Complete the table that relates the pyramid's height to its volume. Then graph the equation $x = \frac{(4V)^{\frac{1}{3}}}{2}$. Where necessary, round values to the nearest tenth.

Volume (V)	Height (x)
0	0
0.25	0.5
2	1
16	2
32	2.5
64	3.2
128	4



3. What is the inverse of the relation represented in the table in Item 2? Complete the table of values.

Height (x)	Volume (V)
0	0
0.5	0.25
1	2
2	16
2.5	32
3.2	64
4	128